ABSTRACT

Few-shot image generation remains challenging because limited training data can reduce image quality, diversity, and semantic consistency. In the Stable Attribute Group Editing (SAGE) framework, the use of the pixel2style2pixel (pSp) encoder can produce latent representations that lose local image details and limit editability during attribute group editing. This study proposes FeatureSAGE, an enhanced few-shot image generation framework that integrates the StyleFeatureEditor inverter and Feature Editor into SAGE. The proposed framework represents images using both W^+ latent codes and intermediate feature tensors, allowing semantic edits to be guided by latent-space directions while feature-level information supports detail preservation during synthesis.

FeatureSAGE was evaluated on the FUNIT Animal Faces and 102 Flowers datasets under a one-shot setting at 1024 $\times$ 1024 resolution. Quantitative evaluation used Fréchet Inception Distance (FID) to measure distributional alignment and Learned Perceptual Image Patch Similarity (LPIPS) to measure perceptual variation. On the 102 Flowers dataset, FeatureSAGE achieved an FID of 45.60, improving over SAGE by 15.8% and over AGE by 25.9%, with an LPIPS of 0.4209. On the Animal Faces dataset, FeatureSAGE achieved an FID of 49.60, improving over SAGE by 18.6% and over AGE by 20.2%, with an LPIPS of 0.4811. These results show that integrating feature-level editing with improved inversion strengthens generated-image distribution alignment while maintaining comparable perceptual variation. The study contributes a FeatureSAGE architecture that extends SAGE through StyleFeatureEditor-based inversion and feature editing for data-scarce generative modeling.

Keywords: Few-shot image generation, StyleGAN2, GAN inversion, attribute group editing, latent space